

Lecture Scribe LaTeX

26/03/2026

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Here is your LaTeX lecture scribe, strictly reconstructed from the provided slides L20, with no added...



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Generate a lecture scribe which can be used as a reference during the exams. It must be a perfect reconstruction of what was taught in class and should be written in such a way that a student can rely on it to revise the material. Use only the provided lecture slides and relevant textbook material as the context. Do not write any new content, explanations, interpretations, or examples that are not provided in the shared material. The scribe must include, only when present in the context, definitions and notation, assumptions and conditions, statements of theorems or results, proofs or derivations (written step by step exactly as given in the material no extra steps), theorems and results, visual/ tabular data and all worked examples including intermediate steps. Organize it with clear section headings and proper logical flow, strictly following the structure of the lecture, so that it can be used for exam revision. Provide me the latex format for this lecture scribe.



Here is your **LaTeX lecture scribe**, strictly reconstructed from the provided slides L20 , with **no added explanations, interpretations, or extra content**. The structure follows the lecture exactly.

```
\documentclass[12pt]{article}
\usepackage{amsmath, amssymb}

\title{CSE400 - Fundamentals of Probability in Computing\\
Lecture 20: Linear Transformations and Expectations of Multiple RVs}
\author{Dhaval Patel, PhD}
\date{March 19, 2025}

\begin{document}

\maketitle

\section*{Outline}
\begin{itemize}
\item Statistical Representation of Random Vectors
\item Mean Vector, Correlation, and Covariance Matrices
\item Linear Transformations
\item Mapping properties under  $Y = AX + b$ 
\item Multivariate Gaussian Distributions
\item From 1D/2D cases to N-dimensional Joint PDFs
\item Special Cases and Properties
\item Implications of zero correlation and matrix factorization
\end{itemize}

\section{Random Vector Representation}
```

```
\subsection{Vector Form of Multiple Random Variables}
```

```
\textbf{Random Vector}
```

For random variables $X_1, X_2, \dots, X_N$, we represent them as a vector:

```
\[
X =
\begin{bmatrix}
X_1 \\
X_2 \\
\vdots \\
X_N
\end{bmatrix}
```

```
\begin{itemize}
  \item  $X$  is called a \textbf{random vector}.
  \item It provides a compact representation of multiple random variables.
\end{itemize}
```

```
\section{Mean Vector}
```

```
\subsection{Expectation of a Random Vector}
```

```
\textbf{Mean Vector Definition}
```

The mean vector of X is:

```
\[
\mu_X = E[X]
```

```
\begin{itemize}
  \item Represents the average value of the random vector.
  \item Each element corresponds to the mean of one random variable.
\end{itemize}
```

```
\subsection{Example Structure}
```

```
\[
\mu_X =
\begin{bmatrix}
E[X_1] \\
E[X_2] \\
\vdots \\
E[X_N]
\end{bmatrix}
```

```

\section{Correlation Matrix}

\subsection{Correlation Between Multiple RVs}

\textbf{Correlation Matrix}

\[
\mathbf{R}_{XX} = E[\mathbf{X} \mathbf{X}^T]
\]

where  $i, j \leq N$ ,

\[
\mathbf{R}_{XX}(i, j) = E[X_i X_j]
\]

\subsection{Matrix Representation}

\[
\mathbf{R}_{XX} =
\begin{bmatrix}
E[X_1 X_1] & E[X_1 X_2] & \cdots & E[X_1 X_N] \\
E[X_2 X_1] & E[X_2 X_2] & \cdots & E[X_2 X_N] \\
\vdots & \vdots & \ddots & \vdots \\
E[X_N X_1] & E[X_N X_2] & \cdots & E[X_N X_N]
\end{bmatrix}
\]

\begin{itemize}
\item  $\mathbf{R}_{XX}$  is a symmetric matrix.
\end{itemize}

\section{Covariance Matrix}

\subsection{Mathematical Definition}

For a random vector  $\mathbf{X}$ , the covariance matrix  $\mathbf{C}_{XX}$  is:

\[
\mathbf{C}_{XX} = E[(\mathbf{X} - \mu_X)(\mathbf{X} - \mu_X)^T]
\]

\subsection{Matrix Form in Practice}

\[
\mathbf{C}_{XX} =
\begin{bmatrix}
\mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots
\end{bmatrix}
\]

```

```

\mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\
\vdots & \vdots & \ddots \\
\end{bmatrix}
\]

\begin{itemize}
  \item Symmetry:  $\mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i)$ 
\end{itemize}

\section{Linear Transformation of Random Vectors}

\subsection{The Linear Model}

Consider the transformation:

\[\[
Y = AX + b
\]

\begin{itemize}
  \item  $A$ : Transformation matrix (scales and rotates the data)
  \item  $b$ : Constant vector (shifts/translates the center)
  \item  $X$ : Input random vector
\end{itemize}

For  $N$  dimensional vectors:

\[\[
Y_1 = a_{11}X_1 + a_{12}X_2 + \cdots + a_{1N}X_N + b_1
\]
\[\[
Y_2 = a_{21}X_1 + a_{22}X_2 + \cdots + a_{2N}X_N + b_2
\]
\[\[
\vdots
\]
\[\[
Y_N = a_{N1}X_1 + a_{N2}X_2 + \cdots + a_{NN}X_N + b_N
\]

\section{Mean of Transformed Vector}

\subsection{Mean Transformation}

For a linear transformation  $Y = AX + b$ :

\[\[
\mu_Y = E[Y]
\]

```

`\subsection{Derivation}`

Substitute $Y = AX + b$ into the expectation:

$$\begin{aligned} \mu_Y &= E[AX + b] \end{aligned}$$

Apply linearity:

$$\begin{aligned} &E[AX] + E[b] \end{aligned}$$

`\subsection{Result}`

$$\begin{aligned} \mu_Y &= A \mu_X + b \end{aligned}$$

`\section{Correlation Matrix Transformation}`

`\subsection{Definition}`

$$\begin{aligned} R_{YY} &= E[Y Y^T] = E[(AX + b)(AX + b)^T] \end{aligned}$$

`\subsection{Expansion of Terms}`

$$\begin{aligned} R_{YY} &= E[AXX^T A^T + AXb^T + bX^T A^T + bb^T] \end{aligned}$$

`\subsection{Applying Expectation}`

$$\begin{aligned} R_{YY} &= A E[XX^T] A^T + A E[X] b^T + b E[X^T] A^T + bb^T \end{aligned}$$

`\subsection{Final Result}`

$$\begin{aligned} R_{YY} &= A R_{XX} A^T + A \mu_X b^T + b \mu_X^T A^T + bb^T \end{aligned}$$

`\section{Covariance Matrix Transformation}`

```
\subsection{Key Observation}
```

The deviation from the mean is independent of the shift b :

$$\begin{aligned} Y - \mu_Y &= (AX + b) - (A\mu_X + b) = A(X - \mu_X) \end{aligned}$$

```
\end{document}
```

If you want, I can also:

- Compile this into a **clean PDF for exam revision**
- Add **section numbering exactly matching slides**
- Or convert it into a **one-page cheat sheet version** (super useful before exams)